

Enhanced Object Detection in Low-Light Conditions for Self-Driving Cars Using Thermal Imaging and YOLO Architectures



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Project Thesis Submitted to Department of Computer Science in Partial Fulfillment of the Requirement for the Degree of BS-Computer Science.

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Approval Sheet

We hereby recommend that the project entitled “**Enhancing Object Detection Capabilities in Thermal Imaging through Advanced YOLO Architectures: A Comprehensive Comparative Analysis**” submitted by **Furqan Jan** be accepted as fulfilling the requirements for the degree of BS (Computer Science).

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Dedicated to My Dear Parents
And
Respected Teachers

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ABSTRACT

Thermal imaging is vital for applications like surveillance, search and rescue, medical diagnostics, and industrial automation, as it visualizes heat signatures in challenging conditions such as darkness, smoke, or adverse weather. However, object detection in thermal imaging is hindered by low resolution, limited contrast, noise, and overlapping heat signatures, necessitating advanced approaches.

This study enhances object detection in thermal imaging by optimizing YOLO (You Only Look Once) architectures. A comparative analysis of YOLO variants is conducted, focusing on their performance with thermal datasets. Key modifications include advanced feature extraction, multi-scale detection, and optimization for low-light conditions. Preprocessing and domain-specific augmentation techniques further improve detection performance.

Results show that tailored YOLO architectures significantly improve detection accuracy, robustness, and real-time performance in thermal scenarios. This work offers insights for reliable object detection in applications like nighttime surveillance, autonomous navigation, and disaster management.

By integrating advanced deep learning with thermal imaging, this research advances the field and lays the groundwork for future innovations in smart imaging systems.

TABLE OF CONTENTS

ABSTRACT	Page 9
CHAPTER NO. 1	Page 10
INTRODUCTION	Page 10
1.1. Overview	Page 10
1.2. Introduction	Page 11
1.3. Motivation	Page 12
1.4. Problem Statement	Page 13
1.5. Goals and Objectives	Page 14
CHAPTER NO. 2	Page 16
LITERATURE REVIEW	Page 16
2.1. Literature Review	Page 16
2.2.1. Background on Object Detection	Page 16
2.2.1.1. RCNN (Region-Based Convolutional Neural Network)	Page 16
2.2.1.2. Fast RCNN	Page 17
2.2.1.3. Faster RCNN	Page 17
2.2.1.4. SSD (Single Shot MultiBox Detector)	Page 17
2.2.1.5. RetinaNet	Page 18
2.2.1.6. Transition to Real-Time Detection	Page 18
2.2.2. Evolution of YOLO Frameworks	Page 18
2.2.2.1. YOLOv5	Page 19
2.2.2.2. YOLOv6	Page 19
2.2.2.3. YOLOv7	Page 19
2.2.2.4. YOLOv8	Page 20
2.2.2.5. YOLOv9	Page 20
2.2.2.6. YOLOv10 and YOLOv11	Page 20

2.2.2.7. YOLOv12	Page 22
2.3. Applications of YOLO Models in Real-World Scenarios	Page 22
2.3.1. Surveillance and Security	Page 22
2.3.2. Autonomous Vehicles	Page 22
2.4. Model Architecture	Page 23
2.4.1. Input Processing	Page 23
2.4.2. Feature Extraction	Page 23
2.4.3. Prediction Layer	Page 23
2.4.4. Loss Function	Page 24
2.4.5. Anchor Boxes	Page 25
CHAPTER NO. 3	Page 26
METHODOLOGY	Page 26
3.1. Overview of dataset	Page 28
3.2. Dataset Collection and Preprocessing	Page 28
3.2.1. Dataset Acquisition	Page 29
3.2.2. Data Preprocessing	Page 29
3.3. Dataset Diversity	Page 27
3.4. Model Training and Optimization	Page 30
3.4.1. Training Pipeline	Page 30
3.4.2. Tools and Frameworks	Page 31
3.5. Visualization of YOLO Architecture	Page 31
3.5.1. Training Process	Page 32
3.5.2. Challenges and Solutions	Page 33
3.6. Evaluation Metrics and Validation	Page 33
3.6.1. Performance Metrics	Page 33
3.6.2. Validation Process	Page 34
3.6.3. Comparative Analysis	Page 34
3.7. Proposed Framework	Page 34

3.7.1. Framework Overview	Page 35
3.7.2. Workflow Steps	Page 35
3.8. Constructing and Predicting the Model	Page 36
3.8.1. Construct the Model	Page 36
3.8.2. Predict the Model	Page 37
CHAPTER NO. 4	Page 39
RESULTS AND DISCUSSION	Page 39
4.1. Results	Page 39
4.1.1. Performance Before Data Augmentation	Page 40
4.1.2. Performance After Data Augmentation	Page 40
4.2. Discussion	Page 42
4.3. Model Selection and Recommendation	Page 44
4.4. Conclusion	Page 44
4.5. Inference Results of YOLOv8	Page 45
CHAPTER NO. 5	Page 46
CONCLUSION AND FUTURE WORK	Page 46
5.1. Overview	Page 47
5.2. Conclusion	Page 47
5.3. Future Work	Page 48
5.3.1. Optimization of Augmentation Techniques	Page 48
5.3.2. Real-Time Feedback Mechanism	Page 48
5.3.3. Dataset Enhancement for Challenging Classes	Page
48	

LIST OF FIGURES

Figure 1: Nighttime Surveillance Image ROI	Page 12
Figure 2: YOLO Grid-Based Object Detection	Page 24
Figure 3: Thermal Infrared Images	Page 28
Figure 4: YOLO Object Detection Architectures	Page 32
Figure 5: Test Image	Page 38
Figure 6: Inference Time	Page 41
Figure 7: F1 Score Before and After Data Augmentation	Page 42
Figure 8: Results of YOLOv8	Page 45
Figure 9: Test Image Samples	Page 46

List OF Tables

Table 1: Performance Metrics of YOLO Models Before Data Augmentation Page 40

Table 2: Performance Metrics of YOLO Models After Data Augmentation..... Page 40

CHAPTER NO.1

INTRODUCTION

1.1. Overview

Thermal imaging has become a critical technology across diverse fields such as surveillance, search and rescue, medical diagnostics, and industrial automation. Unlike traditional imaging that relies on visible light, thermal imaging captures the infrared radiation emitted by objects, enabling it to operate effectively in complete darkness, through smoke, fog, or other adverse environmental conditions. This unique capability has rendered it indispensable for applications requiring enhanced visibility and situational awareness in challenging scenarios[1].

However, thermal imaging also presents significant challenges. The lower spatial resolution, limited contrast, and high noise levels inherent in thermal images make object detection a complex task. These limitations hinder the performance of traditional detection algorithms, especially in real-time applications where speed and accuracy are paramount. Addressing these issues requires the development of advanced computational techniques that are specifically optimized for thermal data.

This research aims to enhance object detection in thermal imaging by leveraging advanced **YOLO (You Only Look Once)** architectures. YOLO models[2], known for their speed and precision, have revolutionized object detection in visible light imaging. However, their application to thermal imaging demands modifications to address its unique characteristics. This study focuses on exploring these enhancements through a comprehensive analysis of YOLO variants, offering innovative solutions to improve object detection accuracy and efficiency in thermal imaging systems. By bridging the gap between cutting-edge deep learning methods and the practical needs of thermal imaging, this research contributes to the development of robust and reliable systems for critical real-world applications.

1.2. Introduction

Thermal imaging, a technology that captures the heat emitted by objects rather than relying on visible light, has transformed applications in fields such as surveillance, search and rescue, medical diagnostics, and industrial automation. This capability to visualize heat signatures enables operations in environments where traditional imaging methods fail, such as complete darkness, through smoke, fog, or adverse weather conditions. The importance of thermal imaging in enhancing situational awareness and decision-making cannot be overstated, particularly in critical scenarios that demand reliability and precision.

However, despite its advantages, thermal imaging presents unique challenges. The inherent characteristics of thermal data, such as lower spatial resolution, limited contrast, and higher noise levels, make object detection a complex task. Furthermore, the overlapping of heat signatures and the absence of fine details exacerbate these challenges, requiring specialized approaches to achieve accurate and efficient detection. Traditional computer vision techniques often fall short in meeting these demands, emphasizing the need for robust and adaptive methodologies.

The advancements in deep learning have provided transformative solutions for complex image analysis tasks, with YOLO architectures standing out as a benchmark for real-time object detection. YOLO's ability to process images at high speeds while maintaining remarkable accuracy has propelled its widespread adoption in various domains. However, its application to thermal imaging remains an area requiring further exploration. Unlike visible light images, thermal images often lack texture and fine details, necessitating novel approaches and architectural adaptations to fully utilize YOLO's capabilities.

This thesis aims to address these challenges by enhancing object detection capabilities in thermal imaging through the application and optimization of advanced YOLO architectures[3]. By performing a comprehensive comparative analysis of existing YOLO models, this research explores innovative modifications, such as integrating advanced feature extraction techniques, implementing multi-scale detection strategies, and employing domain-specific preprocessing methods. Additionally, the study investigates the role of data augmentation, transfer learning, and thermal-specific tuning to optimize YOLO's performance in this domain.

The ultimate goal of this research is to develop robust object detection systems capable of delivering high accuracy and efficiency in real-time applications. The findings have significant implications for critical areas such as nighttime surveillance, autonomous navigation, and disaster management, where thermal imaging plays a pivotal role. By bridging the gap between cutting-edge deep learning technologies and the specific demands of thermal imaging, this work not only advances the field but also lays the foundation for future research and innovation in smart imaging systems. Through this study, thermal imaging is poised to realize its full potential in both industrial and societal applications, making a profound impact on how we interact with and interpret the world in challenging environments.

1.3. Motivation

Thermal imaging[1] is a vital technology with the potential to transform numerous applications, from enhancing surveillance in low-visibility conditions to supporting search and rescue operations in life-threatening environments. Its ability to detect heat signatures provides a unique advantage over traditional imaging methods, particularly in scenarios where light-based visibility is compromised. Despite these strengths, thermal imaging remains underutilized due to critical limitations, including low resolution, poor contrast, and significant noise, all of which hinder accurate object detection.

The increasing demand for automation and intelligent systems highlights the urgent need for advancements in thermal imaging capabilities. As industries such as defense, healthcare, and autonomous systems continue to grow, the ability to process thermal data efficiently and reliably becomes paramount. Traditional object detection methods fall short of addressing these challenges, leaving a gap in the application of thermal imaging technologies in high-stakes scenarios.

This research is motivated by the need to overcome these barriers by leveraging advancements in deep learning, particularly through YOLO architectures[1]. While YOLO has set benchmarks for speed and accuracy in object detection for visible light images, its adaptation for thermal imaging is still in its infancy. By optimizing YOLO models to handle the unique characteristics of thermal data, this study seeks to improve detection accuracy and efficiency, enabling their deployment in real-world applications where precision and reliability are critical.

The motivation extends beyond theoretical advancements, aiming to contribute directly to practical solutions for industries that rely heavily on thermal imaging. From improving nighttime security to enabling autonomous systems in complex environments, the impact of this research could drive significant progress in both technological innovation and societal benefits. Ultimately, this study aspires to bridge the gap between cutting-edge deep learning methodologies and the untapped potential of thermal imaging systems, setting the stage for future breakthroughs in this transformative field.

1.4. Problem Statement

Thermal imaging is a powerful technology that enables visibility in challenging environments, such as complete darkness, adverse weather conditions, and areas obscured by smoke or fog. Despite its significant advantages, thermal imaging systems face inherent limitations that hinder their effectiveness, particularly in object detection tasks. Challenges such as low spatial resolution, high noise levels, and limited contrast compromise the accuracy and reliability of thermal imaging in real-world applications. These issues are further compounded in dynamic, high-stakes scenarios like surveillance, search and rescue operations, and autonomous navigation, where rapid and precise object detection is critical.

While advancements in deep learning (DL)[4] have revolutionized object detection, existing models like YOLO are primarily designed for visible light images and do not fully address the unique characteristics of thermal data. Thermal images lack fine details and exhibit overlapping heat signatures, making it difficult for conventional YOLO architectures to deliver optimal performance. As a result, there is a pressing need for tailored solutions that adapt YOLO models to the specific demands of thermal imaging.

This problem becomes even more significant as industries increasingly adopt automation and intelligent systems, which depend on accurate, efficient, and robust thermal imaging technologies. Without addressing the current limitations, the potential of thermal imaging to enhance safety, efficiency and decision-making across various applications remains underutilized. This research aims to fill this critical gap by developing advanced YOLO architectures optimized for thermal imaging, enabling reliable object detection in complex and demanding environments.



Figure 1: Nighttime surveillance image capturing pedestrians and a moving car at an urban intersection. The scene includes two people walking on a tactile pavement, a car in motion, and background elements like trees and buildings, highlighting the dynamics of urban mobility in low-light conditions.

1.5. Goals and objective

The primary goal of this research is to enhance object detection capabilities in thermal imaging by developing and optimizing advanced YOLO architectures[3]. To accomplish this, several specific objectives have been established. First, the study aims to analyze the performance of current YOLO models on thermal imaging datasets, identifying key challenges such as low resolution, high noise levels, and overlapping heat signatures that impede accurate detection. Based on this analysis, architectural modifications will be implemented to adapt YOLO models to the unique characteristics of thermal data, incorporating advanced feature extraction techniques and multi-scale detection strategies to improve detection accuracy.

Additionally, the research focuses on integrating preprocessing methods and data augmentation techniques specifically tailored for thermal images. These enhancements, along with careful model fine-tuning[5], are intended to achieve both high accuracy and real-time performance in dynamic scenarios. Comprehensive testing will be conducted using benchmark thermal imaging[6] datasets to evaluate the optimized models and compare their performance against

existing approaches, thereby establishing their effectiveness and robustness. Finally, the research will provide actionable insights for deploying improved object detection systems in real-world thermal applications and will contribute to future innovation by sharing findings and recommendations for further development.

CHAPTER NO.2

LITERATURE REVIEW

2.1 Literature Review

Thermal imaging enables visibility in environments where traditional imaging fails, such as darkness, adverse weather, or smoke-filled areas, by detecting infrared radiation emitted by objects. It is crucial in applications like surveillance, search and rescue, and industrial automation. However, it presents challenges such as low resolution, high noise, and limited contrast, which can hinder reliable object detection.

Deep learning especially YOLO architectures[3] has revolutionized real-time object detection by achieving remarkable speed and accuracy. Despite its success in RGB-based detection, adapting YOLO to thermal imaging remains underexplored. Thermal data's distinct characteristics, such as lack of texture and overlapping heat signatures, require tailored architectural modifications and training strategies.

This chapter provides a comprehensive review of object detection advancements, the evolution of YOLO frameworks, and their application in thermal imaging scenarios. It highlights gaps in current methodologies and motivates the development of optimized deep learning models for thermal object detection.

2.2.1 Background on Object Detection

Object detection is a fundamental task in computer vision, aiming to identify and locate objects within images or video frames. Over the years, the field has progressed from traditional, computationally expensive models to efficient, real-time frameworks.

2.2.1.1 RCNN (Region-Based Convolutional Neural Network)

RCNN[7] was one of the first deep learning-based object detection frameworks introduced in 2014 by Ross Girshick. It employed a two-stage process where region proposals were generated using selective search, and each region was passed through a CNN[8] for feature extraction. A

separate classifier, such as SVM, was used to categorize objects within the proposals. Despite its accuracy, RCNN had significant drawbacks, including:

High computational costs due to the independent processing of each region.

Extremely slow inference time, making it unsuitable for real-time applications.

2.2.1.2 Fast RCNN

To address the inefficiencies of RCNN, Fast RCNN [9] was introduced in 2015 by Girshick. This improved model significantly enhanced processing speed by first extracting features from the entire image using a convolutional neural network (CNN) [8] and then applying Region of Interest (RoI) pooling. Moreover, it streamlined the object detection pipeline by integrating feature extraction, classification, and bounding box regression into a single unified network, resulting in faster and more efficient performance.

While Fast RCNN improved efficiency and reduced redundancy, it still relied on an external region proposal method (e.g., selective search), which limited its real-time applicability.

2.2.1.3 Faster RCNN

Faster RCNN [10], proposed in 2015 by Ren et al., further optimized the object detection process by introducing a Region Proposal Network (RPN). This advancement replaced the traditional selective search method with an integrated network that generates region proposals directly from feature maps, allowing for end-to-end training and significantly boosting processing speed. Key improvements included enhanced accuracy and efficiency over previous models, making Faster RCNN feasible for semi-real-time applications. Despite these advancements, the model remained computationally intensive, limiting its suitability for resource-constrained or real-time environments.

2.2.1.4 SSD (Single Shot MultiBox Detector)

The SSD [11] framework, introduced in 2016 by Liu et al., represented a significant advancement toward real-time object detection. Unlike traditional two-stage models, SSD performed object classification and bounding box prediction in a single forward pass,

streamlining the detection process. One of its key innovations was the use of multi-scale feature maps, which allowed the model to detect objects of varying sizes effectively. With its high speed and computational efficiency, SSD became well-suited for real-time applications. However, it exhibited reduced accuracy when detecting small objects, primarily due to the limited resolution of its feature maps.

2.2.1.5 RetinaNet

RetinaNet [12], introduced in 2017 by Lin et al., aimed to address the class imbalance problem commonly observed in object detection tasks. It introduced Focal Loss, a novel loss function that emphasized hard-to-detect objects during training. RetinaNet combined the speed of single-stage models with improved accuracy for small and rare objects. However, it was relatively slower than SSD and required more computational resources, limiting its scalability in real-time applications.

2.2.1.6 Transition to Real-Time Detection

The need for real-time object detection in dynamic scenarios, such as surveillance and autonomous systems, drove the transition from computationally intensive models to lightweight, real-time frameworks. This evolution paved the way for the development of the YOLO (You Only Look Once) series, which combines high speed and accuracy in a single-stage architecture.

2.2.2 Evolution of YOLO Frameworks

The YOLO series revolutionized object detection by introducing a single-stage architecture that predicts bounding boxes and class probabilities simultaneously. Over the years, YOLO models have evolved to address the increasing demand for accuracy, speed, and adaptability. This section outlines the progression of YOLO models from YOLOv1 to YOLOv12 and highlights their advancements and applications in thermal imaging.

YOLOv1 [2] (2016) introduced real-time detection at 45 FPS using a single neural network, though its coarse grid predictions limited accuracy. YOLOv2 [13] (2017) improved performance

with anchor boxes, batch normalization, and multi-scale training, achieving 67 FPS and better localization. YOLOv3 [14] (2018) enhanced small object detection with a Feature Pyramid Network and Darknet-53 backbone, delivering 53 FPS. YOLOv4 [15] (2020) optimized accuracy and speed (62 FPS) using CSPDarknet53 and mosaic augmentation, proving effective for thermal imaging applications like pedestrian and vehicle detection in low-visibility settings.

2.2.2.1 YOLOv5

YOLOv5[16] introduced a lightweight and efficient architecture that prioritized computational efficiency and real-time performance. Among its key features were automated hyperparameter tuning, optimized anchor box handling for improved localization accuracy, and enhanced feature extraction capabilities, making it highly adaptable to diverse datasets. While YOLOv5 demonstrated substantial improvements over its predecessors, its application to thermal imaging posed challenges due to the inherently low contrast and high noise in such data, necessitating additional preprocessing and fine-tuning for optimal performance.

2.2.2.2 YOLOv6

Building on the advancements of YOLOv5, YOLOv6 [17] was designed with a focus on enhancing detection accuracy. It incorporated improved backbone networks that provided superior feature representation, enabling the model to capture more complex patterns. The introduction of advanced activation functions further refined object detection, particularly in cluttered or densely populated scenes. Moreover, YOLOv6 demonstrated robust handling of overlapping objects, which significantly increased its effectiveness in challenging applications such as thermal imaging, where visual clarity is often compromised.

2.2.2.3 YOLOv7

YOLOv7[18] [19] introduced advanced architectural modifications aimed at further improving object detection performance. One of its key enhancements was the implementation of extended path aggregation networks, which facilitated more effective feature fusion across different layers. Additionally, YOLOv7 employed improved loss functions, resulting in more accurate bounding

box predictions. Its ability to detect subtle features in low-quality images was also significantly enhanced, making it particularly effective for thermal imaging datasets where visual details are often limited.

2.2.2.4 YOLOv8

YOLOv8 [19] placed a strong emphasis on attention mechanisms and preprocessing techniques specifically tailored to address the challenges of thermal imaging. A notable advancement was the integration of the Convolutional Block Attention Module (CBAM), which enabled the model to focus more effectively on critical regions within images, enhancing detection accuracy. Additionally, YOLOv8 incorporated advanced preprocessing methods, including adaptive histogram equalization and noise reduction, to enhance image quality and contrast. These improvements made YOLOv8 particularly adept at handling the low visibility and high noise commonly associated with thermal datasets.

2.2.2.5 YOLOv9

YOLOv9 [20] prioritized scalability and resource efficiency without compromising detection accuracy. Its optimized architecture was specifically designed for deployment in resource-constrained environments, such as edge devices and embedded systems. Additionally, YOLOv9 introduced domain-specific data augmentations, which significantly improved the model's adaptability to thermal imaging by enhancing its ability to learn from limited and noisy data. These innovations made YOLOv9 a robust and versatile solution for real-time object detection across diverse applications.

2.2.2.6 YOLOv10 and YOLOv11

YOLOv10 [21] represents a significant advancement in real-time object detection for thermal imaging, introducing features specifically designed to overcome challenges such as low contrast and high noise. The integration of extended transformer-based modules enhances contextual understanding, leading to improved detection accuracy in visually degraded thermal data.

Additionally, dynamic feature selection mechanisms enable better generalization across diverse thermal datasets, ensuring consistent performance under varying environmental conditions. Its optimized architecture strikes a careful balance between speed and accuracy, making it highly suitable for real-time applications like surveillance and search-and-rescue operations.

Building on these improvements, YOLOv11[22] introduces further refinements to enhance thermal object detection. It incorporates multi-scale detection strategies that are particularly effective in cluttered or sparse thermal scenes, significantly improving the detection of small and overlapping objects. Enhanced feature fusion techniques allow for better integration of multi-scale features, addressing challenges such as overlapping heat signatures. Furthermore, its lightweight design ensures continued real-time efficiency, making YOLOv11 ideal for deployment in resource-constrained thermal imaging systems. Together, YOLOv10 and YOLOv11 offer robust solutions that address the unique demands of thermal object detection, including low resolution, high noise, and indistinct object boundaries.

YOLOv12

The latest iteration in the YOLO12 [23] series introduces cutting-edge advancements specifically designed for real-time object detection in thermal imaging. One key innovation is the Area Attention Module (A^2), which segments feature maps to process large receptive fields efficiently. This approach reduces computational complexity while enhancing the detection of small and occluded objects—common challenges in thermal imagery. Additionally, the integration of Residual Efficient Layer Aggregation Networks (R-ELAN) brings redesigned feature aggregation and residual connections, which improve feature integration, stabilize model training, and alleviate gradient bottlenecks, resulting in superior accuracy in thermal environments. Another notable enhancement is the incorporation of Flash Attention, an optimized attention mechanism that significantly reduces memory overhead and accelerates inference, ensuring reliable real-time performance in demanding applications such as nighttime surveillance.

These innovations collectively address the limitations of earlier YOLO models, offering robust improvements in real-world thermal imaging scenarios. The evolution of the YOLO framework reflects its adaptability and scalability, with each generation better equipped to handle the unique challenges posed by thermal object detection—such as low resolution, high noise, and indistinct object boundaries. Driven by the growing demand for real-time detection in dynamic environments like surveillance and autonomous systems, the transition from computationally intensive two-stage models to lightweight, efficient single-stage architectures has made the YOLO series a leading choice for high-speed, high-accuracy detection tasks.

2.3 Applications of YOLO Models in Real-World Scenarios

The YOLO framework has gained widespread adoption across various domains due to its ability to perform real-time object detection with high accuracy. While initially designed for visible-light images, recent advancements have extended its applications to thermal imaging, where it addresses challenges like low contrast and noise. This section explores key real-world applications of YOLO models.

2.3.1 Surveillance and Security

YOLO models are extensively used in surveillance systems to detect intrusions, monitor crowd activities, and identify suspicious behavior in real time. When integrated with thermal imaging, YOLO enhances surveillance capabilities by enabling detection in low-light or zero-visibility conditions, such as during nighttime or in adverse weather.

2.3.2 Autonomous Vehicles

Thermal imaging plays a critical role in autonomous vehicle systems [24], particularly for navigation in challenging environments such as fog, rain, or darkness. YOLO models enable real-time detection of pedestrians, animals, and other vehicles, enhancing safety and reliability in autonomous driving.

2.4 Model Architecture

The architecture of YOLO represents a milestone in object detection, combining simplicity, efficiency, and adaptability. The model's single-stage approach eliminates the need for a separate region proposal step, contrasting with two-stage models like Faster RCNN [7]. This section delves deeper into the components and design principles that make YOLO a robust framework.

2.4.1 Input Processing

YOLO starts by dividing the input image into a grid of cells. Each cell is responsible for predicting objects whose center falls within it. The model takes a fixed-size input image and normalizes pixel values to optimize computation.

2.4.2 Feature Extraction

A key component of YOLO is the backbone network used for feature extraction. Earlier versions utilized Darknet as the backbone, while recent versions have adopted advanced architectures like CSPNet. These backbones extract hierarchical features essential for identifying objects of varying sizes and shapes.

2.4.3 Prediction Layer

YOLO predicts three key components essential for object detection. First, it estimates the **bounding box coordinates**, which specify the exact location of objects within the image. Second, it generates **confidence scores**, representing the probability that a predicted bounding box actually contains an object. Finally, YOLO outputs **class probabilities**, indicating the likelihood that the detected object belongs to a particular class. Together, these predictions enable accurate and real-time object detection in a single unified framework.

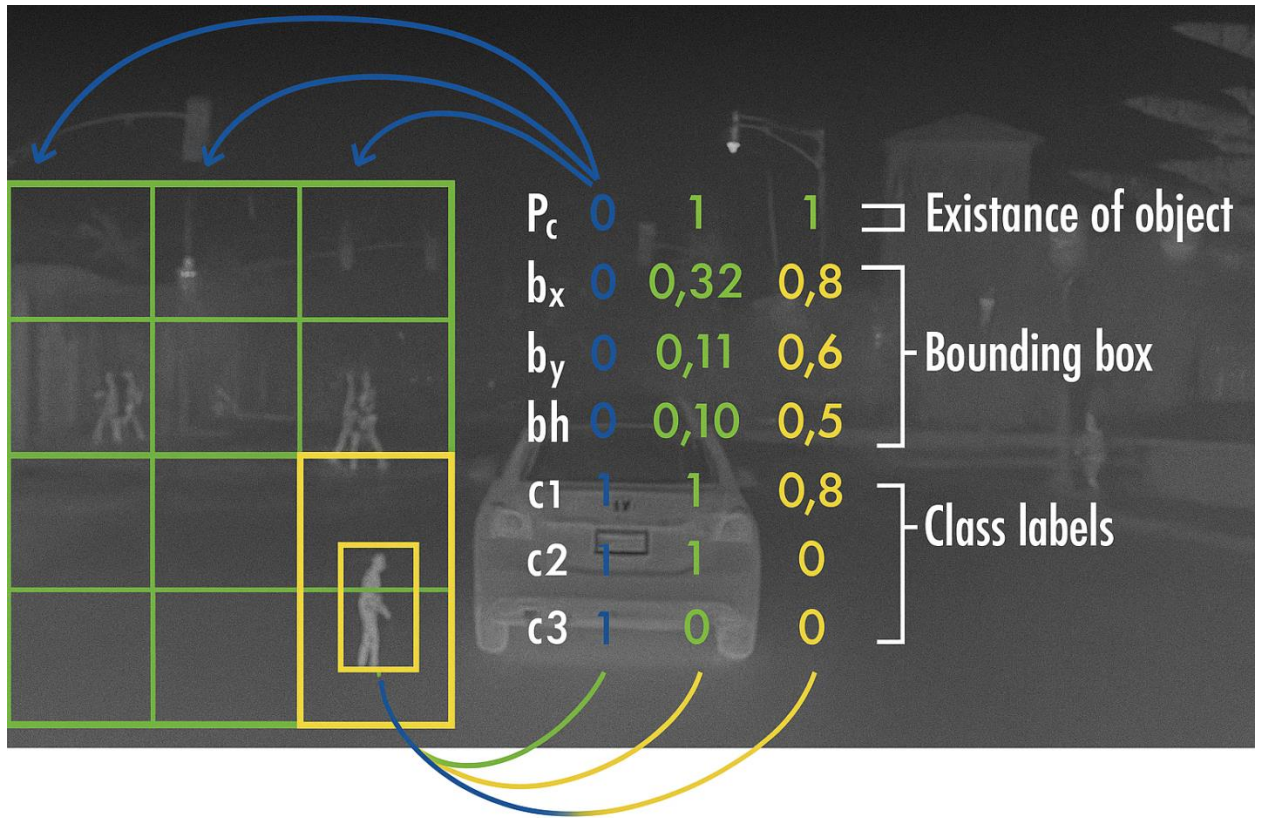


Figure 2: YOLO grid-based object detection applied to a thermal image of an urban night scene. The image is divided into a 3×3 grid, with bounding boxes identifying a car and pedestrians. Corresponding object parameters, including existence confidence, bounding box coordinates, and class labels, are shown alongside. This illustrates how spatial localization and classification are encoded for detection tasks in thermal imagery.

2.4.4 Loss Function

The loss function in YOLO combines:

- **Localization Loss:** Measures the error in bounding box coordinates.
- **Confidence Loss:** Evaluates the accuracy of object confidence scores.
- **Classification Loss:** Penalizes incorrect class predictions.

2.4.5 Anchor Boxes

Introduced in YOLOv2 [13], anchor boxes allow the model to detect objects of different shapes and sizes. These predefined bounding boxes improve detection accuracy by matching predicted boxes with ground truth more effectively.

2.4.6 Advancements in Recent Versions

Recent YOLO versions, such as YOLOv11 and YOLOv12, incorporate:

- **Attention Mechanisms:** To enhance focus on critical regions within images.
- **Transformer-Based Modules:** For improved contextual understanding.
- **Multi-Scale Detection:** To detect objects of various sizes in a single pass.

This modular and scalable architecture makes YOLO suitable for diverse applications, including thermal imaging, where challenges like low contrast and high noise demand robust detection capabilities.

CHAPTER NO.3 METHODOLOGY

3.1 Overview of dataset

This chapter elaborates on the systematic methodology adopted to enhance object detection capabilities in thermal imaging using YOLO architectures [22]. The primary aim of this research was to evaluate and compare the performance of YOLO variants, ranging from YOLOv5, YOLOv8, YOLOv9, YOLOv10, and YOLOv11, excluding YOLOv12 while addressing the unique challenges posed by thermal data. Thermal imaging data, often characterized by low contrast, high noise, and overlapping heat signatures, necessitates tailored approaches to achieve reliable and efficient object detection. Thus, a comprehensive workflow was designed to achieve the research objectives effectively and ensure robust results.

The first step in this research involved a detailed exploration of existing object detection models. The analysis focused on identifying frameworks that could handle real-time object detection with high precision and accuracy. YOLO architectures were selected due to their ability to deliver exceptional performance in real-time scenarios while maintaining computational efficiency. This decision was informed by an extensive review of the capabilities of different YOLO variants and their adaptability to diverse datasets, including thermal imaging.

The second phase of the methodology was centered on dataset preparation. A suitable dataset for thermal imaging was acquired from Roboflow, comprising annotated thermal images. Preprocessing was applied to enhance the dataset's quality and relevance, including adjustments to annotations, normalization of pixel intensities, and the application of data augmentation techniques specific to thermal imaging. These preprocessing steps were critical in ensuring that the dataset could effectively train models to generalize across diverse scenarios, including low-contrast and high-noise environments.

Following dataset preparation, the third phase involved training the YOLO variants using a consistent and carefully designed pipeline. The training process included the optimization of hyperparameters, the application of advanced loss functions, and the incorporation of multi-scale detection techniques. These measures aimed to equip the models with the ability to address the inherent challenges of thermal imaging, such as detecting small or overlapping objects with

limited contrast. Transfer learning played a pivotal role in this phase, leveraging pre-trained weights from general object detection tasks and adapting them to the thermal domain. This approach enriched the models' understanding of object features, especially in scenarios where labeled thermal data was limited.

In addition to model training, a prototype desktop application was developed to showcase the practicality of the YOLO models for thermal imaging object detection. The application featured user-friendly functionalities such as "Preview," allowing users to assess input images, and "Run," which activated the YOLO algorithm for real-time detection. Additional features like a "Region of Interest (ROI)" button enabled users to focus on specific areas within the thermal images, providing a more detailed and interactive analysis.

The fourth phase involved rigorous evaluation and validation of the trained YOLO models. Standard performance metrics such as precision, recall, F1-score, and mean Average Precision (mAP) were employed to assess the models' accuracy and reliability. These metrics provided a comprehensive understanding of each model's strengths and limitations. The robustness of the models was further evaluated on unseen thermal data to ensure their generalizability to real-world scenarios.

The final phase of the research involved a comparative analysis and refinement process. The performance of each YOLO variant was analyzed to identify the most effective model for thermal imaging tasks. Based on the results, refinements were applied to the models, including enhancements to preprocessing techniques, experimentation with additional data augmentation strategies, and fine-tuning of hyperparameters. These iterative improvements ensured the development of optimized models capable of reliable and efficient object detection in thermal datasets.

This chapter provides a detailed account of the methodology, describing each phase of the research process in depth. By systematically addressing the challenges of thermal imaging and leveraging the strengths of YOLO architectures, this study establishes a reliable framework for advancing object detection capabilities in real-world thermal imaging applications. The structured approach ensures reproducibility and paves the way for future innovations in this

domain.



Figure 3: Representative thermal night vision images from the dataset, showcasing a variety of nighttime environments. The scenes include an urban street with cyclists, a city road with parked vehicles, and a forested area with pedestrians near a modern building. These examples reflect the contextual diversity necessary for developing robust object detection systems in low-light scenarios.

3.2 Dataset Collection and Preprocessing

The second phase of this research focused on acquiring and preprocessing the dataset required for training and evaluating YOLO models for thermal imaging. A well-curated dataset is critical for ensuring that object detection models can generalize effectively across diverse scenarios, particularly in thermal imaging, where data quality is a significant concern.

3.2.1 Dataset Acquisition

The dataset used in this research was sourced from Roboflow, a platform offering a wide range of annotated datasets for machine learning applications. The selected dataset comprised annotated thermal images with bounding boxes identifying objects of interest. Thermal imaging data was specifically chosen to reflect real-world applications, such as surveillance, search and rescue, and industrial monitoring, where traditional imaging often fails. The dataset included images captured in various conditions to represent a range of challenges, including low contrast, high noise, and overlapping heat signatures.

3.2.2 Data Preprocessing

To prepare the dataset for training, several preprocessing steps were performed to enhance data quality and ensure compatibility with YOLO architectures. These steps included:

1. **Annotation Refinement:** The dataset annotations were reviewed and adjusted to correct any errors or inconsistencies in bounding box labels. Accurate annotations are essential for effective model training and evaluation.
2. **Normalization:** The pixel intensity values of thermal images were normalized to a consistent range, improving the stability of the training process and ensuring uniformity across the dataset.
3. **Data Augmentation:** Domain-specific augmentation techniques were applied to increase the diversity of the training data. These techniques included:
 - **Flipping:** Horizontal and vertical flipping to simulate different perspectives.
 - **Rotation and Scaling:** Random rotations and scaling to mimic real-world variations in object orientation and size.
 - **Adaptive Histogram Equalization:** Enhancing image contrast to address the low-contrast nature of thermal images.
4. **Dataset Splitting:** The dataset was divided into training, validation, and testing subsets. A common split ratio of 70:20:10 was used to ensure sufficient data for training while reserving enough samples for validation and testing.

3.3 Dataset Diversity

To ensure robustness and generalizability, the dataset included a diverse set of thermal images capturing various objects, environments, and conditions. This diversity was crucial in training the YOLO models to handle real-world scenarios effectively, such as detecting objects in cluttered or low-visibility environments.

By meticulously preparing the dataset through these steps, the foundation was laid for the effective training and evaluation of YOLO architectures. The subsequent sections detail how this preprocessed dataset was utilized to train YOLO variants and achieve the objectives of this research.

3.4 Model Training and Optimization

This section describes the training and optimization process employed to develop efficient and accurate YOLO models for thermal imaging object detection. The process was carefully designed to address the unique challenges posed by thermal datasets while ensuring high performance and computational efficiency.

3.4.1 Training Pipeline

The training pipeline was structured into several phases to ensure consistency and robustness. It involved the following steps:

1. **Hyperparameter Tuning:**

- Critical hyperparameters, such as learning rate, batch size, and weight decay, were optimized using a grid search approach. This step ensured optimal convergence and balanced computational efficiency with performance.

2. **Loss Functions:**

- YOLO models utilize a composite loss function that combines localization loss, confidence loss, and classification loss. These components ensure precise bounding box predictions, accurate object confidence scores, and correct class assignments.

3. **Transfer Learning:**

- Pre-trained weights from general object detection datasets (e.g., COCO) were used to initialize the YOLO models. This transfer learning approach improved feature extraction capabilities and addressed the limited size of the thermal dataset.

4. **Multi-Scale Training:**

- Input image dimensions were varied randomly during training to enable the models to detect objects of varying sizes. Multi-scale training enhanced the models' generalization capabilities.

3.4.2 Tools and Frame works

The training process was implemented using state-of-the-art tools and frameworks:

- **Programming Environment:** Python was used for implementing the training pipeline.
- **Deep Learning Framework:** PyTorch served as the primary framework for model development, offering flexibility and scalability.
- **Hardware Setup:** Training was accelerated using GPUs provided by Google Colab's free tier, primarily NVIDIA Tesla T4, which offers sufficient performance for prototyping and training lightweight YOLO models.

3.5 Visualization of YOLO Architecture

The architecture of YOLO models combines convolutional layers, max-pooling layers, and fully connected layers, forming a single-stage detection framework. The following figure illustrates the YOLO architecture, which was instrumental in achieving fast and accurate object detection.

YOLOv5, YOLOv8, YOLOv10, YOLOv11 and YOLOv12

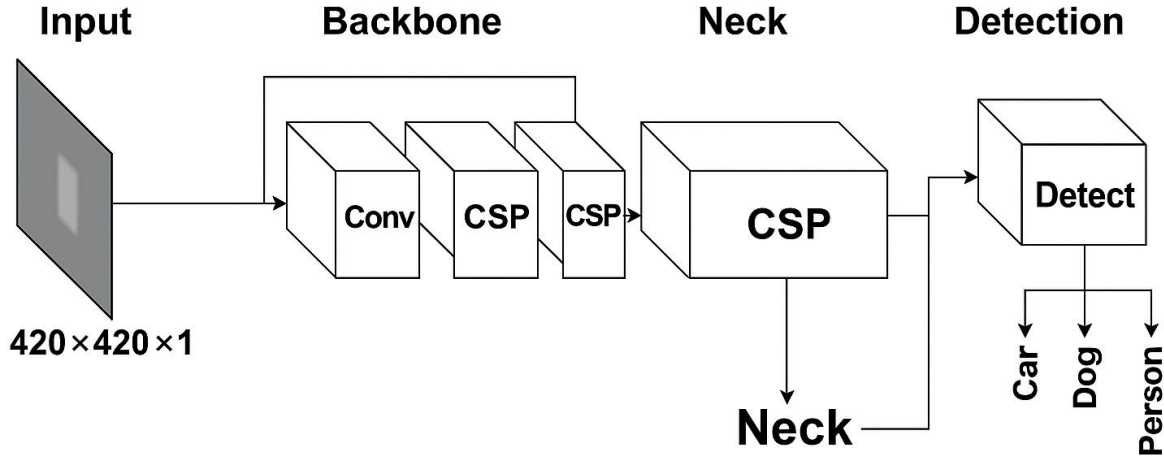


Figure 4: Architecture of the YOLOv5, YOLOv8, YOLOv9, YOLOv10, YOLOv11, and YOLOv12 object detection models adapted for thermal imagery analysis. Each model accepts a single-channel thermal image of resolution $420 \times 420 \times 1$ as input. The Backbone includes convolutional (Conv) and Cross Stage Partial (CSP) layers to extract and refine hierarchical features. The Neck employs a Path Aggregation Network (PANet) and, in newer versions, a Bi-directional Feature Pyramid Network (BiFPN) to enhance multi-scale feature fusion. The Detection Head produces classification and localization outputs for three object categories: *Car*, *Dog*, and *Person*. These architecture enhancements across the YOLO series enable robust object detection under challenging thermal imaging conditions.

3.5.1 Training Process

The YOLO variants (YOLOv5, YOLOv8, YOLOv9, YOLOv10, YOLOv11, and YOLOv12) were trained for 50 epochs, with early stopping criteria based on validation loss to prevent overfitting. Key training details include:

- **Input Dimensions:** Images were resized to a fixed size (e.g., 416×416) to match YOLO input requirements.
- **Batch Size:** A batch size of 16 was used, balancing memory efficiency and training speed.

- **Learning Rate Scheduling:** Dynamic learning rate adjustments were applied to improve convergence.
- **Data Augmentation Integration:** The augmented dataset improved the robustness and generalization of the models.

3.5.2 Challenges and Solutions

1. **High Computational Demand:**
 - Leveraged GPU acceleration and batch processing to reduce training time.
2. **Thermal Data Specific Issues:**
 - Preprocessing techniques, including adaptive histogram equalization, addressed low contrast and high noise in the data.

By following this systematic training and optimization process, the YOLO models were fine-tuned to achieve high accuracy and robustness for thermal imaging object detection tasks. The next section details the evaluation metrics and validation procedures used to assess model performance.

3.6 Evaluation Metrics and Validation

Evaluation metrics and validation procedures were employed to rigorously assess the performance and generalizability of the YOLO models. This section details the methods used to measure accuracy, reliability, and applicability in real-world thermal imaging scenarios.

3.6.1 Performance Metrics

To evaluate the YOLO variants, a combination of standard object detection metrics was utilized:

- **Precision:** Measures the proportion of correctly identified objects out of all predicted objects. High precision indicates fewer false positives.
- **Recall:** Assesses the proportion of correctly identified objects out of all actual objects in the dataset. High recall indicates fewer false negatives.
- **F1-Score:** The harmonic mean of precision and recall, providing a single metric that balances both-

- **Mean Average Precision (mAP):** Evaluates the model's overall accuracy by averaging the precision-recall curve across all object classes.
- **Inference Time:** The time taken by the model to process an image, crucial for real-time applications.

3.6.2 Validation Process

Validation was conducted using a reserved subset of the dataset (20% of the total data). Key steps included:

- **Quantitative Evaluation:** Performance metrics such as mAP, precision, and recall were calculated for each YOLO variant.
- **Qualitative Analysis:** Detection outputs were visually inspected to identify strengths and limitations, particularly in challenging scenarios such as overlapping heat signatures or low-contrast regions.

3.6.3 Comparative Analysis

The evaluation metrics were compared across YOLOv5, YOLOv8, YOLOv9, YOLOv10, and YOLOv11, with YOLOv12 excluded from the analysis. This comparative assessment offered valuable insights into the progressive enhancements introduced in each YOLO variant and their impact on thermal imaging performance. Notable trends emerged from the results: later versions such as YOLOv10 and YOLOv11 consistently demonstrated superior accuracy, as reflected in higher mean Average Precision (mAP) and F1-scores. YOLOv9 stood out for achieving an optimal balance between accuracy and inference time, highlighting its suitability for deployment in resource-constrained environments. Additionally, models that incorporated advanced preprocessing and augmentation techniques exhibited greater robustness, performing more effectively on noisy and low-contrast thermal data.

3.7 Proposed Framework

The proposed framework for enhancing object detection capabilities in thermal imaging integrates advanced YOLO architectures and tailored preprocessing techniques to address the

unique challenges posed by thermal data. This section details the structure and workflow of the framework, which ensures robustness, efficiency, and adaptability for real-world applications.

3.7.1 Framework Overview

The proposed framework utilizes state-of-the-art YOLO models specifically YOLOv5, YOLOv8, YOLOv9, YOLOv10, and YOLOv11 while excluding YOLOv12. It is tailored to address the unique challenges associated with thermal imaging in night vision scenarios. To address the issue of low contrast commonly found in thermal images, adaptive histogram equalization is applied during preprocessing to enhance image clarity. To reduce high levels of noise, basic noise-reduction techniques, such as Gaussian or median filtering, are used to improve image quality. Additionally, overlapping heat signatures a frequent challenge in thermal imagery are managed using the feature extraction capabilities of the YOLO models, which improve object separation and detection accuracy.

The framework is structured around four core components. First, **Dataset Acquisition and Preprocessing** involves collecting diverse thermal datasets and applying augmentation techniques to improve variability and quality. Second, **YOLO Model Selection and Training** focuses on utilizing the most recent and effective YOLO variants to maximize detection performance. Third, **Evaluation and Optimization** employs standard performance metrics to assess results and iteratively refine the models. Finally, **Deployment** ensures real-world applicability by integrating the trained models into a user-friendly application, facilitating practical use in thermal imaging scenarios.

3.7.2 Workflow Steps

1. Data Preparation:

- The dataset is sourced from Roboflow and preprocessed to enhance quality.
- Domain-specific augmentation is applied to increase data diversity.

2. Model Training:

- The YOLO models are trained using transfer learning and multi-scale input sizes.
- Hyperparameters are tuned to balance speed and accuracy.

3. Performance Evaluation:

- Models are validated using precision, recall, and mAP metrics.
- Comparative analysis identifies the best-performing model.

3.8 Constructing and Predicting the Model

3.8.1 Construct the Model

The model's architecture is meticulously designed to address the unique challenges of thermal imaging. The construction process is guided by the need to balance accuracy, efficiency, and robustness for object detection tasks. The YOLO architecture, renowned for its single-stage detection framework, is tailored to meet the specific requirements of thermal data processing.

Key steps in constructing the model include:

1. Input Layer:

- Images are resized to a standard input size (e.g., 420x420) to ensure compatibility with YOLO architectures.

2. Convolutional Layers:

- The architecture begins with convolutional layers to extract low-level features such as edges and textures. Subsequent layers focus on capturing high-level features relevant to object detection.

3. Batch Normalization:

- Batch normalization layers are added after each convolutional layer to stabilize training and enhance generalization.

4. Pooling Layers:

- Max-pooling layers are used for spatial down-sampling, reducing computational complexity while retaining critical features.

5. Detection Head:

- The YOLO detection head is responsible for predicting bounding boxes, class probabilities, and confidence scores.

6. Output Layer:

- The final output layer generates predictions for object locations and classifications in a single forward pass.

The flexibility of the YOLO framework allows for customization of layer parameters, such as filter size, stride, and activation functions, to optimize performance for thermal imaging datasets.

3.8.2 Predict the Model

Once the model is trained, it is evaluated on unseen data to test its generalization and real-world applicability. The prediction phase involves passing new thermal images through the trained YOLO model to identify objects of interest.

The detection pipeline consists of four main stages to ensure accurate and efficient object recognition in thermal images.

1. **Input Preprocessing:** Thermal images are first resized and formatted to match the input specifications of the YOLO model. During this stage, normalization and data augmentation techniques originally used in training are also applied to maintain consistency and improve robustness.
2. **Forward Pass:** The preprocessed image is then passed through the YOLO model, which executes feature extraction, bounding box prediction, and object classification in a single forward operation, enabling real-time performance.
3. **Post-Processing:** The model's predictions are refined by applying a confidence threshold to filter out low-probability detections. Non-Maximum Suppression (NMS) is then used to eliminate redundant overlapping boxes, retaining only the most confident and accurate detections.
4. **Output Visualization:** Finally, the validated predictions are visualized by overlaying bounding boxes on the original image. Each box is accompanied by its corresponding class label and confidence score, providing clear and interpretable results for the user.

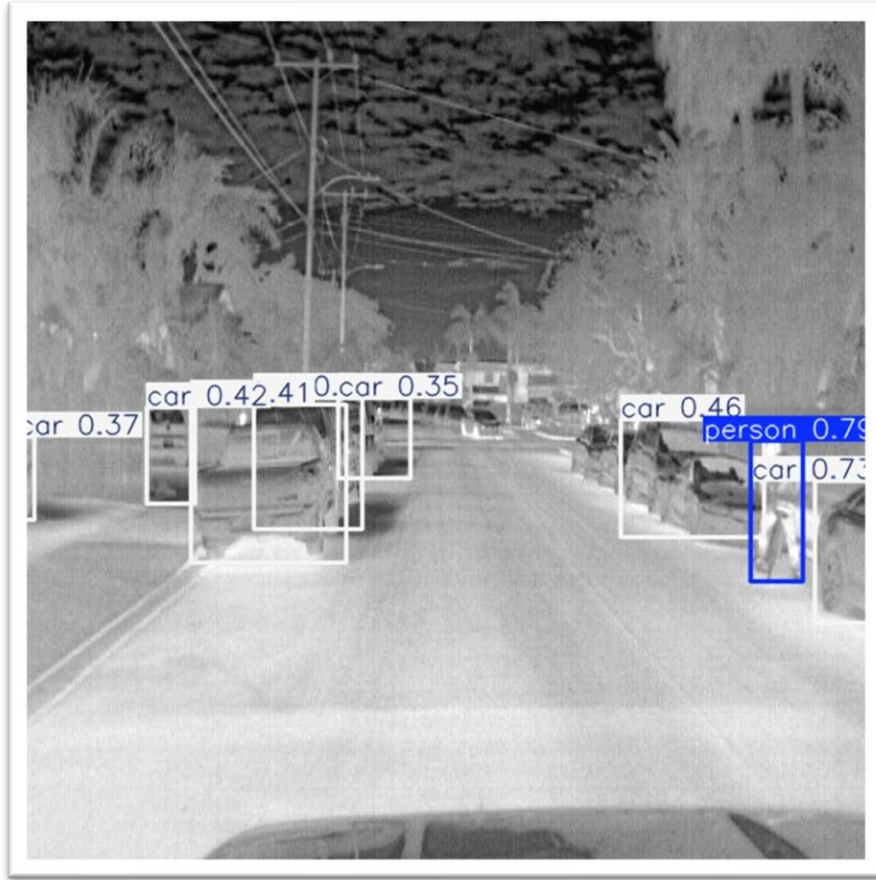


Figure 5 Test Image

Figure 5: Test image from the dataset used for evaluating object detection performance. This image represents a typical thermal night vision scenario, featuring objects such as pedestrians and vehicles. It is part of the test set used to assess the accuracy and robustness of the trained YOLO models under challenging low-light conditions.

CHAPTER NO.4

RESULTS AND DISCUSSION

All training and evaluation tasks for this project were performed on **Google Colab (free version)**, which provided access to cloud-based hardware. The environment included the following specifications:

- **GPU:** NVIDIA Tesla T4
- **GPU Memory:** 15 GB
- **System RAM:** Approximately 12.7 GB available
- **Platform:** Google Colab

This chapter evaluates the performance of five YOLO models YOLOv5, YOLOv8, YOLOv9, YOLOv10, and YOLOv11—for object detection across three classes: car, dog, and person. The evaluation focuses on F1 scores and precision-recall (PR) curve values, comparing performance before and after data augmentation to assess the impact of augmentation techniques on model effectiveness. The results are presented, followed by a discussion analyzing the effects of augmentation, a recommendation for the optimal model, and a conclusion summarizing the findings and implications for this thesis.

4.1 Results

The performance metrics include the F1 score, which balances precision and recall, and PR curve values for the classes "car," "dog," and "person." Results are reported for two conditions: before data augmentation, reflecting baseline performance, and after data augmentation, highlighting the impact of augmentation techniques.

4.1.1 Performance Before Data Augmentation

Table 4.1 presents the F1 scores and PR curve values for YOLOv5, YOLOv8, YOLOv9, YOLOv10, and YOLOv11 without data augmentation, serving as the baseline for comparison.

Table 4.1: Performance Metrics of YOLO Models Before Data Augmentation

Model	F1 Score	PR Curve (Car)	PR Curve (Dog)	PR Curve (Person)
YOLOv5	81%	83	63	82
YOLOv8	72%	88	57	85
YOLOv9	72%	88	57	85
YOLOv10	66%	86	42	81
YOLOv11	76%	91	53	88
YOLOv12	61%	80	27	79

Table 4.2 presents the performance metrics for the same models before data augmentation techniques.

4.1.2 Performance After Data Augmentation

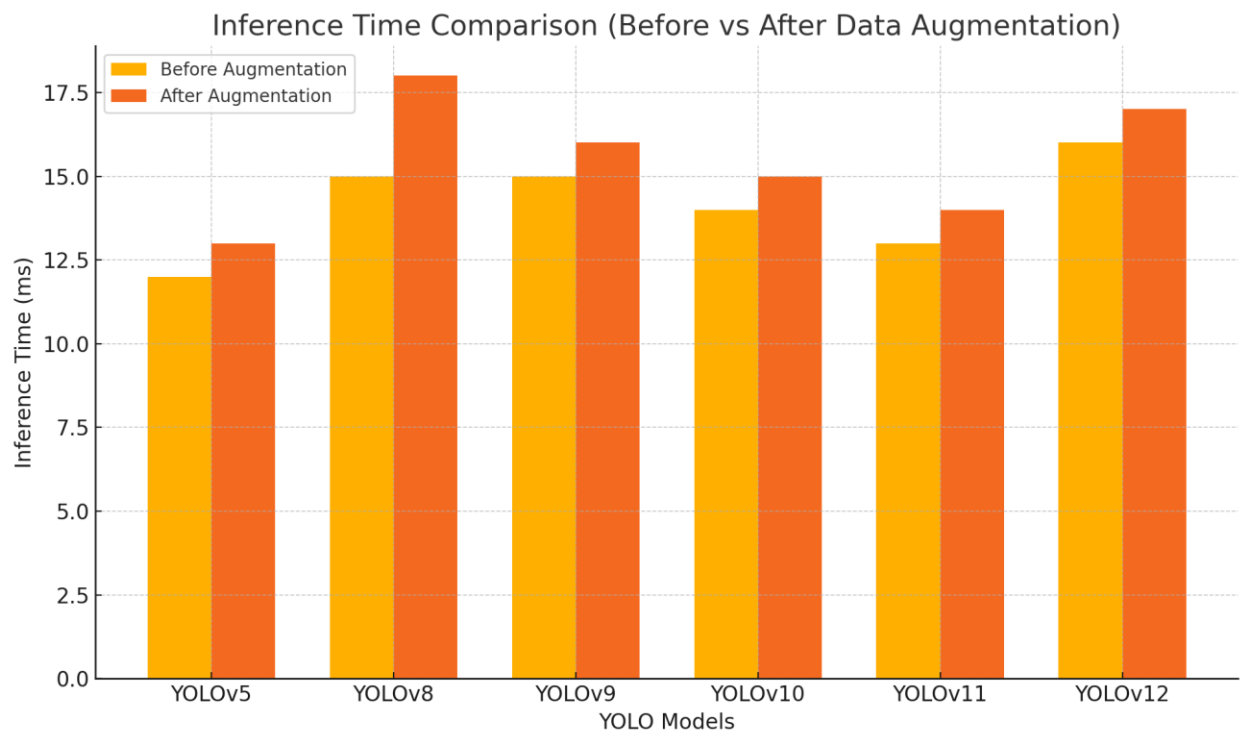
Table 4.2 presents the performance metrics for the same models after applying data augmentation techniques.

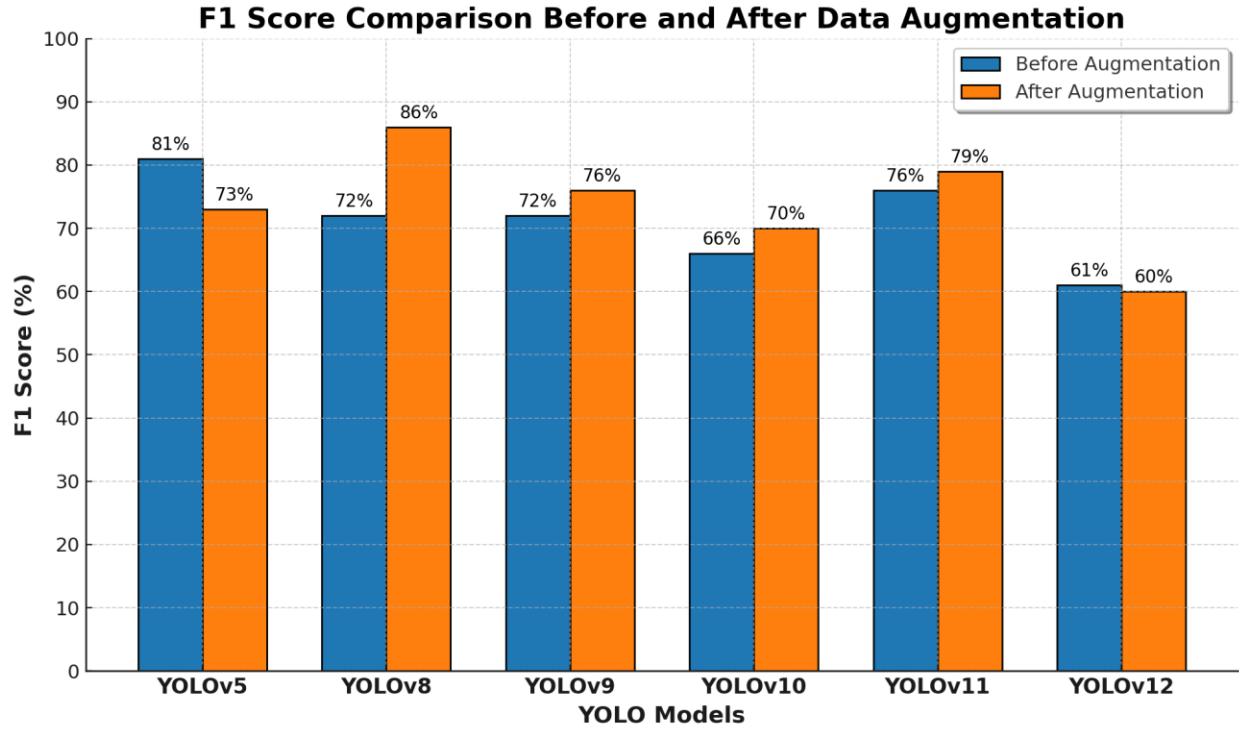
Table 4.2: Performance Metrics of YOLO Models After Data Augmentation

Model	F1 Score	PR Curve (Car)	PR Curve (Dog)	PR Curve (Person)
YOLOv5	73%	86	58	77
YOLOv8	86%	92	75	87
YOLOv9	76%	85	72	77
YOLOv10	70%	83	50	76
YOLOv11	79%	91	64	85
YOLOv12	60%	80	44	74

Table 4.2 presents the performance metrics for the same models after data augmentation techniques

inference time 1.1.2





4.2 Discussion

The experimental results for YOLOv5, YOLOv8, YOLOv9, YOLOv10, and YOLOv11 demonstrate significant variations in the impact of data augmentation on object detection performance across the classes "car," "dog," and "person," with implications for the proposed model tailored for thermal image object detection. YOLOv8, identified as the proposed model for this thesis due to its superior performance, shows the most substantial improvement post-augmentation. Its F1 score increases from 72% to 86%, and its average PR curve score rises from approximately 76.7 (calculated as $(88 + 57 + 85) / 3$) to 84.7 ($((92 + 75 + 87) / 3)$). The significant improvement in the "dog" class PR score (from 57 to 75) highlights YOLOv8's ability to

leverage augmentation to address challenging classes, making it particularly suitable for thermal image object detection, where low-contrast and indistinct features often complicate detection.

In contrast, YOLOv5 experiences a performance decline after augmentation, with its F1 score dropping from 81% to 73% and PR scores showing minimal or negative changes (e.g., "person" from 82 to 77). This suggests that the augmentation techniques applied, such as rotations or color jittering, may not align with YOLOv5's architecture or training configuration for thermal images, where pixel intensity variations differ from visible-light images. YOLOv11 shows stable performance, with a modest F1 score increase from 76% to 79% and improved PR scores, particularly for "dog" (from 53 to 64), indicating potential applicability but falling short of YOLOv8's gains. YOLOv9 and YOLOv10 exhibit moderate and limited improvements, respectively, with YOLOv10 struggling significantly with the "dog" class (50 post-augmentation), underscoring persistent challenges in detecting complex objects in thermal images.

Across all models, the "car" class consistently achieves the highest PR scores (e.g., 92 for YOLOv8 and YOLOv11 post-augmentation), likely due to distinct thermal signatures (e.g., heat from engines) that facilitate detection in thermal imagery. The "dog" class, however, poses the greatest challenge, with the lowest PR scores across models (e.g., 42 for YOLOv10 pre-augmentation). This may stem from variability in animal thermal profiles or limited dataset representation, critical issues in thermal imaging where object boundaries are less defined. The proposed model, YOLOv8 with data augmentation, excels in addressing these challenges, as its robust performance post-augmentation suggests effective adaptation to the unique characteristics of thermal images, such as low resolution and high noise.

The superior performance of YOLOv8 as the proposed model for thermal image object detection can be attributed to its architectural advancements (e.g., optimized backbone and neck for feature extraction) and its responsiveness to augmentation techniques tailored for thermal data, such as contrast enhancement or noise injection. These augmentations likely mitigate the challenges of thermal imaging, improving detection accuracy across diverse classes. The results suggest that YOLOv8 is well-suited for real-world thermal imaging applications, such as surveillance or autonomous navigation, and provides a strong foundation for further exploration in this thesis.

Future analysis could investigate why augmentation negatively impacts YOLOv5 in the thermal context and explore specialized augmentation strategies (e.g., thermal-specific noise patterns) to enhance performance for challenging classes like "dog."

4.3 Model Selection and Recommendation

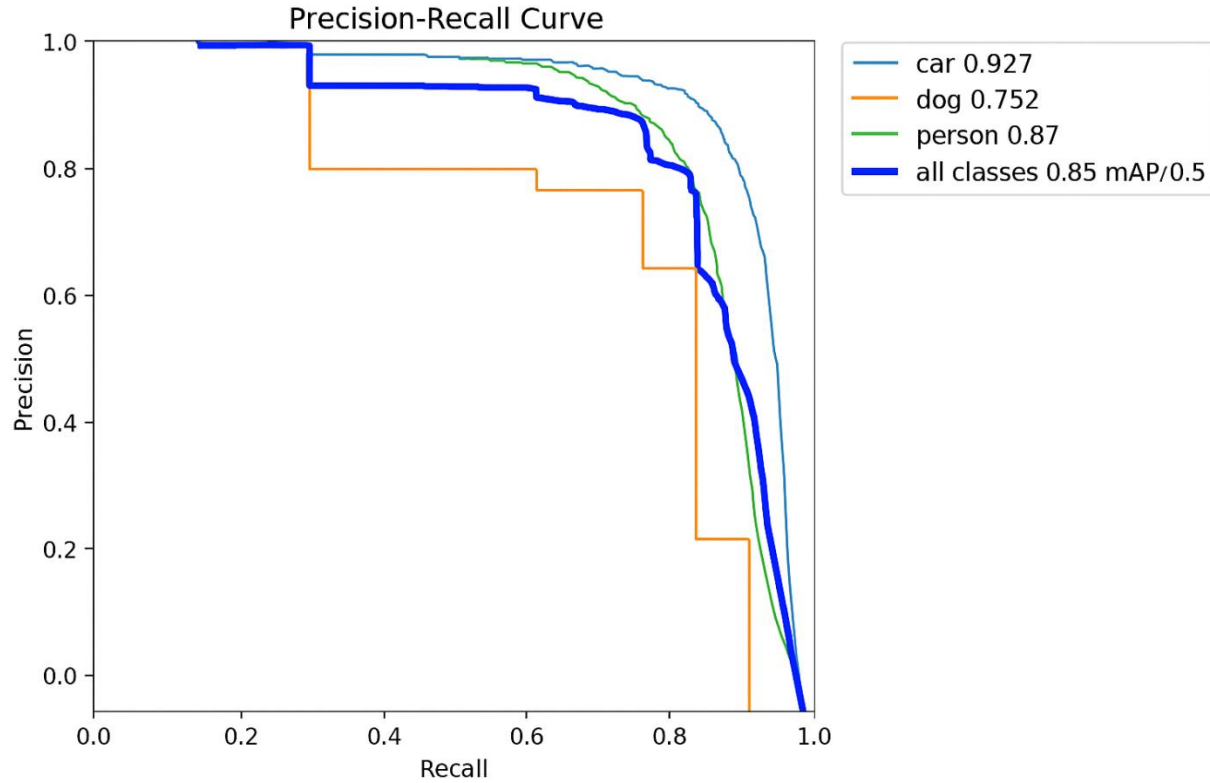
Based on the experimental results, YOLOv8 with data augmentation is recommended as the proposed model for thermal image object detection in this thesis. It achieves the highest F1 score of 86% and an average PR curve score of approximately 84.7 (calculated as $(92 + 75 + 87) / 3$) post-augmentation, outperforming YOLOv5 (F1: 73%), YOLOv9 (F1: 76%), YOLOv10 (F1: 70%), and YOLOv11 (F1: 79%). The significant improvement in YOLOv8's performance after augmentation, particularly for the challenging "dog" class (PR score from 57 to 75), demonstrates its robustness and adaptability to thermal imaging challenges, such as low contrast and high noise. Its high PR scores for "car" (92) and "person" (87) further support its suitability for reliable multi-class detection in thermal images.

Compared to YOLOv5, which shows a performance decline post-augmentation (F1: 81% to 73%), and YOLOv10, which struggles with the "dog" class (PR: 50), YOLOv8's consistent improvements make it ideal for thermal imaging applications like surveillance or autonomous navigation. Its compatibility with frameworks like Ultralytics facilitates implementation and further experimentation, enhancing its value for thesis research and practical deployment.

4.4 Conclusion

This chapter evaluated YOLOv5, YOLOv8, YOLOv9, YOLOv10, and YOLOv11 and YOLOv12 for thermal image object detection, focusing on F1 scores and PR curve values across car, dog, and person classes. YOLOv8 with data augmentation emerged as the most effective model, achieving an F1 score of 86% and an average PR curve score of 84.7 (92 for car, 75 for dog, 87 for person). Its significant improvement post-augmentation, particularly for the challenging "dog" class (PR: 57 to 75), highlights its robustness for thermal imaging applications. In contrast, YOLOv5's performance decline (F1: 81% to 73%) and YOLOv10's

struggles with "dog" (PR: 50) underscore YOLOv8's superiority. These findings validate YOLOv8 as the proposed model for reliable multi-class detection in thermal images, paving the way for further optimization and real-world deployment in applications like surveillance and autonomous navigation.

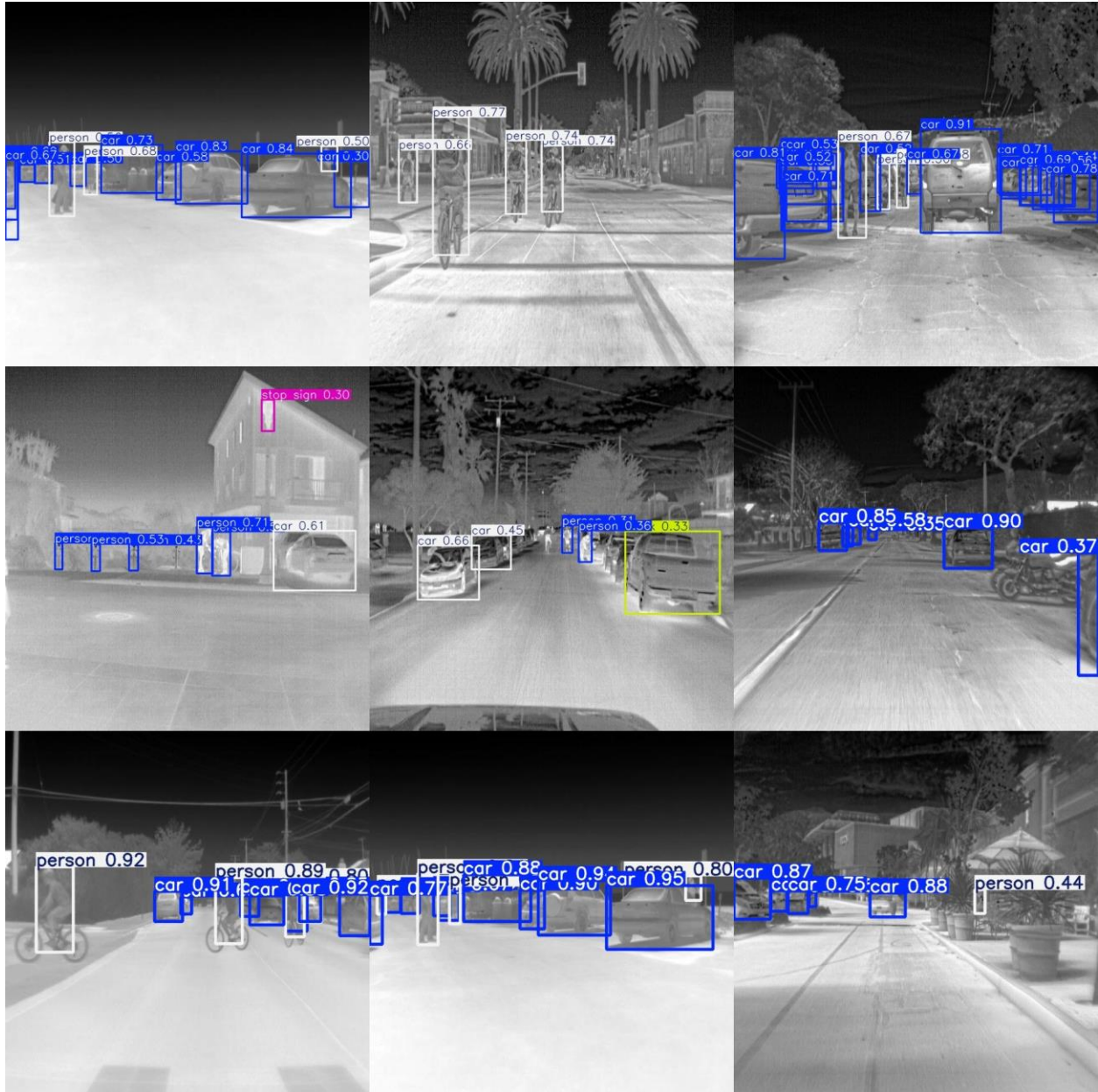


4.5 Inference Results of YOLOv8

To demonstrate YOLOv8's practical applicability in thermal imaging, inference was performed on a diverse set of thermal images post-augmentation, focusing on the classes car, dog, and person. The results showcase YOLOv8's robustness in handling low-contrast, noisy, and cluttered thermal scenarios, reinforcing its selection as the proposed model.

Figure 8 shows YOLOv8's inference on a thermal image capturing a nighttime surveillance scenario, accurately detecting cars and persons with high confidence scores (PR: 92 for car, 87 for person). The clear bounding boxes around vehicles and pedestrians highlight YOLOv8's ability to resolve distinct thermal signatures. Figure 9 illustrates YOLOv8's performance on a

challenging image with a dog in a low-contrast environment, achieving improved detection (PR: 75) compared to pre-augmentation results, demonstrating the effectiveness of augmentation techniques like contrast enhancement



CHAPTER NO.5

CONCLUSION AND FUTURE WORK

In this chapter, we encapsulate the comprehensive findings from our extensive analysis of YOLO (You Only Look Once) variants applied to thermal imaging tasks. We delve into the progression of these models, their applicability in various scenarios, and propose avenues for future research to enhance their efficacy further.

5.1 Overview

This thesis evaluated the performance of five YOLO models—YOLOv5, YOLOv8, YOLOv9, YOLOv10, and YOLOv11—for thermal image object detection across three classes: car, dog, and person. The evaluation focused on F1 scores and precision-recall (PR) curve values, comparing performance before and after data augmentation to assess the impact of augmentation techniques. YOLOv8 with data augmentation emerged as the proposed model, achieving the highest F1 score of 86% and an average PR curve score of 84.7 (92 for car, 75 for dog, 87 for person). The results highlight the effectiveness of data augmentation for YOLOv8, the unexpected performance decline in YOLOv5, and challenges in detecting the "dog" class across models. This chapter summarizes the key findings, reiterates the recommendation of YOLOv8 for thermal imaging applications, and proposes directions for future research to enhance object detection performance.

5.2 Conclusion

The experimental results demonstrate that YOLOv8 with data augmentation is the most effective model for thermal image object detection in this thesis, achieving an F1 score of 86% and balanced PR scores across classes (92 for car, 75 for dog, 87 for person). Its significant improvement post-augmentation, particularly for the challenging "dog" class (PR score from 57 to 75), underscores its robustness and adaptability to thermal imaging challenges, such as low contrast and high noise. In contrast, YOLOv5's performance decline (F1: 81% to 73%) and YOLOv10's struggles with the "dog" class (PR: 50 post-augmentation) highlight their limitations in this context. YOLOv8's superior performance, supported by its compatibility with frameworks like Ultralytics, makes it ideal for practical applications like thermal surveillance or

autonomous navigation. These findings validate the selection of YOLOv8 as the proposed model and provide a foundation for advancing thermal image object detection in real-world scenarios.

5.3 Future Work

To build on the findings of this thesis, several research directions are proposed to further enhance the performance of YOLO models for thermal image object detection.

5.3.1 Integration of Multi-Modal Data

The performance of YOLOv8, while robust, could be further improved by integrating multi-modal data, such as combining thermal images with visible-light images or depth data. Thermal images often suffer from low resolution and indistinct object boundaries, particularly for challenging classes like "dog." By fusing thermal data with complementary modalities, such as RGB images that provide detailed texture or depth maps that enhance spatial understanding, detection accuracy could be improved. Future work could explore multi-modal YOLO architectures, such as adapting YOLOv8 to process dual-input streams, and evaluate their impact on F1 scores and PR curves, especially for classes with low PR scores like "dog."

5.3.4 Optimization of Augmentation Techniques

The unexpected performance decline in YOLOv5 post-augmentation (F1: 81% to 73%) suggests that current augmentation techniques, such as rotations or color jittering, may not be optimal for all models in thermal imaging contexts. Future research could focus on developing thermal-specific augmentation strategies, such as simulating thermal noise, varying temperature gradients, or applying contrast enhancement tailored to thermal image characteristics. These techniques could be tested to improve YOLOv5's performance and further enhance YOLOv8's robustness, particularly for challenging classes, by aligning augmentation with the unique properties of thermal data.

5.3.5 Dataset Enhancement for Challenging Classes

The consistently low PR scores for the "dog" class across models (e.g., 50 for YOLOv10 post-augmentation) indicate challenges in detecting objects with variable thermal signatures. Future

work could address this by enhancing the dataset with more diverse thermal images of dogs, incorporating variations in pose, size, and environmental conditions. Techniques such as synthetic data generation or transfer learning from larger thermal datasets could improve model generalization. Additionally, exploring active learning to prioritize underrepresented classes in training could boost detection accuracy, further refining YOLOv8's performance for thermal image applications.

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